2 INPhO-2007

1. (a) $\vec{p} = m(a\hat{r} + r\omega\hat{\theta})$
   $\vec{F} = m(-r\omega^2\hat{r} + 2a\omega\hat{\theta})$

   (b) $\Delta W = \dfrac{m\omega^2 r^2}{2}$

   (c) Trajectory will be a spiral.

2. (a) $\theta_m = \dfrac{\cot^{-1}(2\mu)}{2}$

   (b) Possible range of θ_m : $\theta_m \in]0,\, \pi/4[$

3. (a) $t \approx 222$ s

   (b) $t = 166$ s

   (c) Process (3a) takes more time since as the disk heats up, its specific heat also increases and more heat is required to effect a further rise in temperature.

4. (a) $d^2 = R^2 \left(1 - \dfrac{(n^2-2)^2}{4}\right)$

   (b) The allowed range of n is $n \in]\sqrt{2}, 2[$

5. (a) $E(r) = \dfrac{\lambda}{2\pi\epsilon_0 r}$

   (b) $V(r) = -\dfrac{\lambda}{2\pi\epsilon_0} \ln\left(\dfrac{r}{a}\right)$

   (c) $\left(x - d\dfrac{k^2+1}{k^2-1}\right)^2 + y^2 = \dfrac{4d^2k^2}{(k^2-1)^2}$ This is an equation of a circle with centre at $\left(d\dfrac{k^2+1}{k^2-1}, 0\right)$ and radius given by $\dfrac{2dk}{|k^2-1|}$. The constant k is related to the "equipotential" V_0.

   (d) See Fig. (2) and part (c)

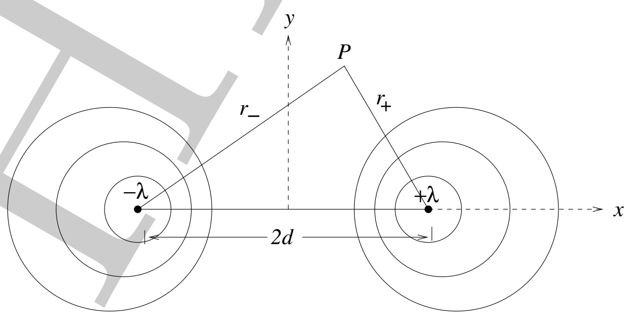

Figure 2: Problem 5 (d)

   (e) $v = 1/\sqrt{\epsilon_0\mu_0} = c$

6. (a) $F = \dfrac{\mu_0 I_1 I_2}{\sqrt{3}\pi}\left[\dfrac{\sqrt{3}S}{2a} - \ln\left(1 + \dfrac{\sqrt{3}S}{2a}\right)\right]$ and direction downwards.

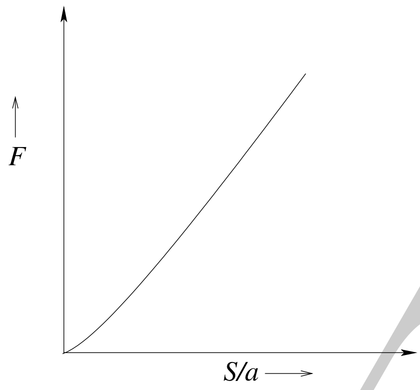

Figure 3: Problem 6 (b)

(b) See Fig. (3).

7. (a) $|\vec{F}| = 2.82 \times 10^{-7}$ N
   (b) $\sigma = 7.27 \times 10^{2}\,\mathrm{C{\cdot}m^{-2}}$
   (c) Energy density after 10 s$= 2.93 \times 10^{16}\,\mathrm{J{\cdot}m^{-3}}$
   (d) The range of kinetic energy of electrons is from 0 to 0.16 eV.

8. (a) $\Gamma = -\,\dfrac{(\gamma-1)}{\gamma}\dfrac{m_a g}{R}$
   (b) $\Gamma = -0.01\ \mathrm{K{\cdot}m^{-1}}$ (i.e. 1^0 C decrease for every 100 m)
   (c) $p = p_0\left(\dfrac{T_0 - \Gamma\, z}{T_0}\right)^{m_a g/R\Gamma}$
   (d) Substituting the given values in the above equation, the height of the atmosphere is approximately 30 km.

9. (a) $B = \dfrac{\mu_0}{2}\dfrac{N_f}{R_f} I(t)$ and direction given by right hand thumb rule.
   (b) $|\varepsilon| = \dfrac{\mu_0}{2}\dfrac{N_p N_f}{R_f}\pi {R_p}^2 \dfrac{dI(t)}{dt}$
   (c) $N_p N_f = 645$
   (d) $M = \dfrac{\mu_0}{2}\dfrac{N_p N_f}{R_f}\pi {R_p}^2 = 1.59 \times 10^{-5}\,\mathrm{H}$
   (e) $N_f = 18$ turns, $N_p = 36$ turns
   (f) Induced emf in case of
      i. Iron : will increase.
      ii. Wood : no appreciable change.
      iii. Copper: decrease.